

Exercise 2.1. Suppose n unrelated people are gathered together. What is the smallest n for which chances are $>50\%$ that there will be two or more people born in the same calendar month?
"At least one": Use Complement

Find the smallest n such that: $P(\text{At least two people born in the same month}) \geq 0.5$

(X) Direct Calculation?

- $P(\text{two people share a month})$
 - $P(\text{exactly three})$
 - $P(\text{four people share same month})$
 - $P(\text{two pairs in different months})$...
- Too many ways...

(✓) Complement Calculation?

- $1 - P(\text{No one shares same month}) = P(\text{At least one match})$
"n" people all have different months

- Calculating $P(\text{No one shares same month})$:

P'	$1 - P'$
$n=1: P: \frac{12}{12} = 1$	0 (0%)
$n=2: P: \frac{12}{12} \cdot \frac{11}{12} \approx 0.9167$	≈ 0.0833 (8.33%)
$n=3: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \approx 0.7639$	≈ 0.2361 (23.61%)
$n=4: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \cdot \frac{9}{12} \approx 0.5729$	≈ 0.4271 (42.7%)
$n=5: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \cdot \frac{9}{12} \cdot \frac{8}{12} = 0.3819$	≈ 0.6181 (61.8%)

$n=5$

The smallest number of people such that the chance of at least two sharing the same birth month exceeds 50%

Exercise 3.1. A fair die is rolled twice. What is the probability that at least one of the two rolls is a six if the sum of the two rolls is at least 10?

$P(B) = \text{All Possible Combinations} : 6$

Two rolls w/ sum ≥ 10

- (6, 4)
 - (4, 6)
 - (6, 5)
 - (5, 6)
 - (6, 6)
- (5, 5) $\times \rightarrow$ Doesn't contain 6
- $P(A \cap B) = 5$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = 5/6$$

- A: at least one of rolls a '6'.
- B: the sum of two rolls ≥ 10 .

Exercise 3.4. Suppose $P(A) = P(B) = .9$. Give a useful lower bound on $P(B|A)$.

Inclusion-Exclusion Formula
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Conditional Probability
 $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq 1$$

↓ rearrange

$$P(A \cap B) \geq P(A) + P(B) - 1$$

flip sign

Plug in $\circ P(A) = 0.9$
 $\circ P(B) = 0.9$

$$P(A \cap B) \geq 0.9 + 0.9 - 1$$

$$P(A \cap B) \geq 0.8$$

- No matter what, the p (both events happening) can't be lower than 0.8.
- Overlap is at least 0.8.

Apply to Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\geq 0.8}{0.9}$$

$$P(B|A) \geq \frac{8}{9} \quad \text{Lower Bound for } P(B|A)$$

Exercise 3.10. Three independent proofreaders will read a manuscript. The probabilities that a particular error will be detected by them are respectively .92, .85, and .95. What is the probability that the error will go undetected?

$$\begin{aligned} A_1 &: \{ \text{Proofreader 1 detecting error} \} \\ A_2 &: \{ \text{Proofreader 2 detecting error} \} \\ A_3 &: \{ \text{Proofreader 3 detecting error} \} \end{aligned} \Rightarrow P(A_1 \cap A_2 \cap A_3) \rightarrow \text{All proofreaders detecting error}$$

$$\begin{aligned} A_1^c &: \{ \text{Proofreader 1 misses error} \} \\ A_2^c &: \{ \text{Proofreader 2 misses error} \} \\ A_3^c &: \{ \text{Proofreader 3 misses error} \} \end{aligned} \Rightarrow \begin{aligned} & \text{⊗ } P(A_1^c \cap A_2^c \cap A_3^c) \neq 1 - P(A_1 \cap A_2 \cap A_3) \rightarrow \text{At least one misses error.} \\ & \text{⊙ } P(A_1^c \cap A_2^c \cap A_3^c) = [1 - P(A_1)] \cdot [1 - P(A_2)] \cdot [1 - P(A_3)] \end{aligned}$$

$$P(A_1^c \cap A_2^c \cap A_3^c) = (1 - 0.92)(1 - 0.85)(1 - 0.95)$$

$$P(A_1^c \cap A_2^c \cap A_3^c) = 0.0006$$

Probability all proofreaders miss error

Exercise 3.11. * Two numbers are chosen without replacement from $1, 2, \dots, 100$.

What is the probability that the smaller number is greater than 20 if the larger number is smaller than 60?

Conditioned on larger number

Let: X be smaller number
 y be larger number

• Possible X and y ranges to satisfy $X < y$:

$X: \{1 \dots 99\}$

$y: \{2 \dots 100\}$ Total of 99 possible values each.

• Possible X ranges to satisfy $X > 20$:

$X: \{21 \dots 99\}$

Total of 79 possible values.

$X < y < 60$

• Possible y ranges to satisfy $y < 60$:

$y: \{2 \dots 59\}$

Total of 58 possible values. (for y only)

Accounting for X , additional $X=1$

Total of 59 possible values $X < y < 60$

• Count $\binom{59}{2} = 1711$

$20 < X < y < 60$

• Possible X and y ranges to satisfy $X > 20 \cap y < 60$:

$X: \{21 \dots 99\}$

$y: \{2 \dots 59\} \Rightarrow$ Overlapping: $\{21 \dots 59\}$

Total of 39 possible values. $20 < X < y < 60$

• Count: $\binom{39}{2} = \frac{39 \cdot 38}{2} = 741$

Combination: order is not important

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

★ **Equilikely Outcomes:** In an equally finite sample space, the probability:

$$P(E) = \text{Favorable Outcomes Count} / \text{Total Outcomes Count}$$

$$P(X > 20 | y < 60) = \frac{P(X > 20 \cap y < 60)}{P(y < 60)}$$

$$P(X > 20 | y < 60) = \frac{\overset{\text{count}}{|X > 20 \cap y < 60|} / \binom{100}{2}}{|y < 60| / \binom{100}{2}}$$

$\Downarrow$

$$P(X > 20 | y < 60) = \frac{\text{Count of } X > 20 \cap y < 60}{\text{number of } (x, y) \text{ pairs}} / \text{Count of } y < 60$$

$$P(X > 20 | y < 60) = 741 / 1711 = 0.4332$$